

Q1

Solve the problem with CPU scheduling algo's.

Draw a table with 4 rows with 3 columns having the following details

Process	AT	BT	Content switch is 5 sec.
P_1	0	5	
P_2	0	4	
P_3	0	3	

apply: (i) SRTF (ii) Priority scheduling.

(ii) FCF5 (ii) SJF (iii) Roundrobin scheduling

solve w CT, Gantt chart, TAT, WT, Avg TAT, Avg WT.

$$TAT = CT - AT$$

$$WT = TAT - BT$$

(i) FCF5

order $\rightarrow P_1 \rightarrow P_2 \rightarrow P_3$

Gantt chart

$$|P_1|^{5}|CS|^{5}|P_2|^{4}|CS|^{4}|P_3|^{3}|CS|^{3}$$
Calculation:

Process	AT	BT	CT	TAT	WT
P_1	0	5	5	5	0
P_2	0	4	14	14	10
P_3	0	3	22	22	19

$$\text{Avg WT} = \frac{0 + 10 + 19}{3} = 9.67$$

$$\text{avg TAT} = \frac{5 + 14 + 22}{3} = 13.67$$

(ii) SJF $\rightarrow$ start with shortest BT: $P_3(3) \rightarrow P_2(4) \rightarrow P_1(5)$ order: $P_3 \rightarrow P_2 \rightarrow P_1$

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Gantt chart:- $P_3 \mid CS \mid P_2 \mid CS \mid P_1$

calculation

Process	AT	BT	CT	TAT	WT
P_1	0	5	22	22	17
P_2	0	4	12	12	8
P_3	0	3	13	3	0

$$\text{avg WT} = \frac{17 + 8 + 0}{3} = 8.33$$

$$\text{avg TAT} = \frac{22 + 12 + 3}{3} = 12.33$$

ii) SRIF

order: $P_3 \rightarrow P_2 \rightarrow P_1$

Gantt chart:

$P_3 \mid CS \mid P_2 \mid CS \mid P_1$

calculation:

Process	AT	BT	CT	TAT	WT
P_1	0	5	22	22	17
P_2	0	4	12	12	8
P_3	0	3	3	3	0

$$\text{avg WT} = 8.33 \quad \text{avg TAT} = 12.33$$

iv) priority scheduling: can't solve cause values not given.

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Q. (i) SRTF is suitable because it selects the task with the shortest remaining CPU burst time

(ii) $\rightarrow 0-3$: MP runs

$$\text{remaining MP} = 20 - 3 = 17$$

$\rightarrow$ At 3: TF arrives with $BT = 1$

TF is shorter than MP $\rightarrow$ TF runs from 3-4

$\rightarrow 4-5$: MP continues

$$\text{remaining MP} = 16$$

$\rightarrow$ At 5: WN arrives with $BT = 2$

WN is shorter than MP $\rightarrow$ WN runs.

$\rightarrow$ At 6: CL arrives with $BT = 4$, but WN has only 1 unit remaining, so WN continues.

$\rightarrow 5-7$: WN completes

$\rightarrow 7-11$: compare MP = 16 and CL = 4 $\rightarrow$ CL wins.

$\rightarrow 11-27$: CL finishes, so remaining MP runs until completion.

$\rightarrow$ Gantt chart:

MP³ TF¹ MP⁵ WN² CL⁴ MP²²

$\rightarrow$ Final execution order

MP $\rightarrow$ TF $\rightarrow$ MP $\rightarrow$ WN $\rightarrow$ CL $\rightarrow$ MP

(iii) Gives quick response to short tasks - such as touch events, notifications & camera launch

$\rightarrow$ SRTF improves responsiveness by prioritizing short remaining tasks

(iv) response time $\Rightarrow$ First CPU start time - AT

Task	AT	BT	Time CPU start	CT	TAT	WT	RT
MP	0	20	0	27	27	7	0
TF	3	1	3	4	7	0	0
WN	5	2	7	9	4	0	0
CL	6	4	7	11	5	7	7

$$TAT = CT - AT \quad ; \quad WT = TAT - BT$$

→ MP, TF and WN all have RT = 0 so, music plays, touch event and whatsapp notification share the fastest response time of 0 seconds.

Q3 (i) starvation

→ occurs when a process waits for a very long time because other processes are repeatedly selected for execution

ex: a low priority process keeps waiting while high priority process continuously executes.

(ii) conveyor effect: occurs in FCFS scheduling when a long process makes many short process wait behind it.

ex: a long process runs first, causing several short processes to wait

Context switching: saving the current process's state and loading another process's state so the CPU can switch between them.

Aging: Gradually increasing the priority of processes that have waited for a long time to prevent starvation.

Q4) i) FCFS

Gantt chart:

⁰P₁ ⁴P₂ ¹³P₃ ¹⁹P₄ ²³P₅ ³²P₆ ³⁷P₇ ⁴²P₈ ⁵¹P₉ ⁵⁷

Process	AT	BT	CT	TAT	WT	RT
P ₁	0	4	4	4	0	0
P ₂	0	4	13	13	4	4
P ₃	0	6	19	19	13	13
P ₄	0	4	23	23	19	19
P ₅	0	9	32	32	23	23
P ₆	0	6	38	38	32	32
P ₇	1	4	42	41	37	37
P ₈	1	9	51	50	41	41
P ₉	1	6	57	55	49	49

avg: TAT = 30.56 WT = 24.22

(ii) SJF: once a process starts, it finishes completely.

order: P₁ → P₄ → P₇ → P₃ → P₆ → P₉ → P₂ → P₅ → P₈

Gantt chart:

⁰P₁ ⁴P₄ ⁸P₇ ¹²P₃ ¹⁸P₆ ²⁴P₉ ³⁰P₂ ³⁹P₅ ⁴³P₈ ⁵⁷

Process	AT	BT	CT	TAT	WT	RT
P ₁	0	4	4	4	0	6
P ₄	0	4	8	8	4	4
P ₇	1	4	12	11	7	7
P ₃	0	6	18	18	12	12
P ₆	0	6	24	24	18	18
P ₉	2	6	30	28	22	22
P ₂	0	9	39	39	30	30
P ₅	0	9	43	43	39	39
P ₈	1	9	57	56	47	47

$$\text{Avg TAT} = 26.22$$

$$\text{Avg WT} = 19.89$$

(iii) SRTF

Execution:

$$\rightarrow 0-4 = P_1$$

$$\rightarrow 4-8 = P_4$$

$$\rightarrow 8-12 = P_2$$

$$\rightarrow 12-18 = P_3$$

$$\rightarrow 18-24 = P_4$$

$$\rightarrow 24-30 = P_9$$

$$\rightarrow 30-39 = P_2$$

$$\rightarrow 39-48 = P_5$$

$$\rightarrow 48-57 = P_8$$

Gantt chart: $P_1 \overset{0}{|} P_4 \overset{4}{|} P_2 \overset{8}{|} P_3 \overset{12}{|} P_6 \overset{18}{|} P_9 \overset{24}{|} P_2 \overset{30}{|} P_5 \overset{39}{|} P_8 \overset{48}{|} P_1 \overset{57}{|}$

Process	CT	TAT	WT	RT
P_1	4	4	0	0
P_4	8	8	4	4
P_2	12	11	7	2
P_3	18	19	12	12
P_6	24	24	18	18
P_9	30	28	22	22
P_2	39	39	30	30
P_5	48	48	39	39
P_8	57	56	42	42

$$\text{Avg TAT} = 26.22 \quad \text{Avg WT} = 19.89$$

(iv) priority scheduling

Order: $P_2 \overset{9}{|} P_6 \overset{19}{|} P_8 \overset{27}{|} P_3 \overset{35}{|} P_6 \overset{39}{|} P_4 \overset{45}{|} P_1 \overset{49}{|} P_4 \overset{53}{|} P_2 \overset{57}{|}$

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Process	AT	BT	CT	TAT	WT	RT
P ₁	0	9	9	9	0	0
P ₅	0	9	9	18	9	9
P ₈	1	9	27	26	17	17
P ₃	0	6	33	33	27	27
P ₆	0	6	39	39	33	33
P ₉	2	6	45	43	37	37
P ₇	0	4	49	49	45	45
P ₄	0	4	53	52	49	49
P ₂	1	4	57	56	52	62